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The homogeneous Poisson point process

The Poisson process has a central role in point process statistics. It is fundamental to any successful analysis of point pattern data that the user is familiar with the basic properties of this process.

There are many situations in which the Poisson process is a suitable model, i.e. when the points are ‘randomly’ distributed in space. In addition, it serves as a basis for the construction of more complicated models. Perhaps even more importantly, the Poisson process is a null or benchmark model that may be used as a reference model to distinguish between point patterns exhibiting aggregation and repulsion. Finally, it admits analytical calculations of summary characteristics and thus provides some understanding of the theory of point processes.

Therefore, this chapter presents the theory and application of the Poisson process systematically and in detail. The exposition starts with the binomial point process, which is a finite variant of the Poisson process and provides an easy introduction into its theory. Then the Poisson process is defined and thoroughly investigated. A detailed justification is given as to why the Poisson process is the ideal model for

complete spatial randomness. The simulation of the Poisson process is easy to understand, but the derivation of the summary characteristics is a little technical in places. However, the resulting formulas are valuable tools in applications as they allow a comparison of empirical characteristics with theoretical ones and facilitate the general understanding of the various summary characteristics used throughout this book.

Finally, issues of statistical inference are discussed – in particular, tests of the CSR hypothesis. Testing this hypothesis is a central problem in point process statistics. Indeed, when analysing a spatial point pattern one aims to detect and quantify spatial structure, with the final aim of relating it to underlying processes that have caused the observed pattern. Hence, it is necessary to verify that an observed aggregation or repulsion of points is really significant. Rejection of the CSR hypothesis for a given pattern can indicate this and may then be followed by the more complicated but also more interesting part of the point process analysis.

2.1 Introduction

This chapter introduces the simplest and most important infinite point process model: the homogeneous Poisson point process. More complicated models are much easier to understand once this null model has been discussed in detail. In addition, the Poisson process plays a central role in the theory of point processes as a reference or null model, leading to direct statistical applications.

A specific point pattern may well exhibit various types of interaction between its constituent points. For instance, the points may occur in clusters (see Figures 1.4, 1.9, 6.3 and 6.4) or may exhibit regularity (see Figures 1.2, 1.6, 6.7 and 6.23). In addition, there may be a *hard-core distance*, i.e. a disc or sphere of diameter r_0 around each point where no other points are located. In other words, a positive minimal inter-point distance may be determined for a given pattern. All of the above features may occur in combination in the same pattern. If neither of the above interactions is found in a point pattern, it may be regarded as completely spatially random. This property of a pattern is termed *complete spatial randomness* (CSR). A theoretical model for patterns with this property forms an important basis for comparison, as a *null model* and as a reference for the construction of summary characteristics.

Furthermore, calculations may be carried out of the extent and probability of fluctuations from the theoretical characteristics in samples. It is thus possible to determine objectively whether a specific fluctuation in an observed pattern is small enough to be considered insignificant, as it is a feature that might well be observed in a pattern that is a realisation from a completely random point process model.

By imposing axioms concerning stationarity and lack of interaction that are intuitively appealing, completely random processes can be characterised as Poisson point processes. As a result, it is possible to decide whether a given pattern exhibits spatial clustering or regularity or whether it may be considered CSR.

In addition, the Poisson process has a further role as a *basic building block for other more complicated models*; see Chapter 6. The definition of many cluster processes is based on Poisson processes, some hard-core processes result from Poisson processes by thinning and Gibbs processes are defined with direct reference to Poisson processes. Therefore, simulation procedures for point processes frequently comprise the construction of a Poisson point process, which is then modified to yield the required form.

For the interested reader, a history of the concept of the Poisson process may be found in Daley and Vere-Jones (1988, 2003). They report that the first recorded use of the Poisson process in spatial statistics appears to be that of Abbe (1879) as discussed in Section 1.3.2.

2.2 The binomial point process

2.2.1 Introduction

The *binomial point process* is the classical starting point for a discussion of CSR and hence of the Poisson process. However, while intuitively appealing, the binomial process does not suffice as a model for CSR. Nevertheless it is interesting as the most simple non-trivial example of a spatial point process model, and the Poisson process may then be considered to be a generalisation of the binomial process. Furthermore, it is a null model for finite point processes; see Chapter 3.

A binomial point process consists of n points, which are randomly scattered in a set W . The term ‘randomly’ means here that the n points $x_1, \dots, x_n$ are uniformly and independently distributed in W . The set W is assumed to be bounded. Its area is denoted by $\nu(W)$, where the neutral symbol ‘ ν ’ may also denote volume, if W is a subset of $\mathbb{R}^3$, i.e. in the context of a three-dimensional spatial pattern.

Consider first a single point randomly distributed in space. Whereas this yields a very simple example of a trivial one-point pattern of no direct practical relevance, the union of several of these points yields the binomial point process. More specifically, a single random point x is *uniformly distributed* in W if

$$\mathbf{P}(x \in A) = \frac{\nu(A)}{\nu(W)} \quad (2.2.1)$$

for all subsets A of W . This equation means that the probability that x takes its position in the subset A of W is equal to the ratio of the areas of the sets A and W . This is the classical definition in the sense of geometrical probability.

Consider now a more interesting point process which consists of n points and apply formula (2.2.1) for each of these points. This yields the binomial point process for which

$$\begin{aligned}\mathbf{P}(x_1 \in A_1, \dots, x_n \in A_n) &= \mathbf{P}(x_1 \in A_1) \cdot \dots \cdot \mathbf{P}(x_n \in A_n) \\ &= \frac{\nu(A_1) \cdot \dots \cdot \nu(A_n)}{\nu(W)^n},\end{aligned}\quad (2.2.2)$$

where $A_1, \dots, A_n$ are subsets of W . This implies that all points are randomly scattered in W and are independent of each other's locations.

Figure 2.2 (on p. 64 below) shows the result of a simulation of a binomial point process in $W = \square$ with $n = 50$ points.

If the points $x_1, \dots, x_n$ form a binomial point process in W then the random pattern formed by these points is denoted $N_{W(n)}$. The ordering of the points is ignored and $N_{W(n)}$ can be regarded as a random set.

The binomial point process $N_{W(n)}$ is the first non-trivial example of a point process in this book. It is close to but not in every sense *the* model for CSR, as will be shown below: in spite of the independence assumption in the construction there are some spatial correlations in the pattern, which result from the fact that the total number of points in W is fixed, i.e. equal to n . Only by assuming a random number of points and extending the approach to the whole space can the right definition of CSR be given.

2.2.2 Basic properties

The binomial point process $N_{W(n)}$ owes its name to a distributional property. If A is a subset of W then the random number of points in A , denoted by $N_{W(n)}(A)$, follows a binomial distribution, with parameters $n = N_{W(n)}(W)$ and $p = p(A) = \nu(A)/\nu(W)$. More specifically:

$$\mathbf{P}(N_{W(n)}(A) = k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for } k = 0, \dots, n. \quad (2.2.3)$$

Since the mean (or expected value) of a binomial distribution is np , the mean number of points in A is

$$np = n \frac{\nu(A)}{\nu(W)} = \lambda \nu(A),$$

where λ is the mean number of points per unit area or volume, termed the *intensity* of the binomial point process,

$$\lambda = \frac{n}{\nu(W)}.$$

In general, void probabilities are very important characteristics for point processes. They are concerned with the event that a given set K is empty, i.e. that it does not contain any point, and are given for the binomial point process by

$$\mathbf{P}(N_{W(n)}(K) = 0) = \frac{(\nu(W) - \nu(K))^n}{\nu(W)^n}, \quad (2.2.4)$$

where $\mathbf{P}(N_{W(n)}(K) = 0)$ is the probability that there is no point in the subset K of W .

In addition, some of the finite-dimensional distributions of the binomial point process are given by a simple formula: if $A_1, \dots, A_k$ are disjoint subsets of W with $A_1 \cup \dots \cup A_k = W$ and if $n_1 + \dots + n_k = n$, then

$$\begin{aligned} \mathbf{P}(N_{W(n)}(A_1) = n_1, \dots, N_{W(n)}(A_k) = n_k) \\ = \frac{n!}{n_1! \cdot \dots \cdot n_k!} \cdot \frac{\nu(A_1)^{n_1} \cdot \dots \cdot \nu(A_k)^{n_k}}{\nu(W)^n}. \end{aligned} \quad (2.2.5)$$

This distribution is the well-known multinomial distribution with parameters n , $p_1 = \frac{\nu(A_1)}{\nu(W)}$, $\dots$, $p_n = \frac{\nu(A_n)}{\nu(W)}$, which is familiar from the analysis of count data using log-linear models. For overlapping sets $A_1, \dots, A_k$ the formula is more complicated and will not be considered here.

Note that numbers of points in different subsets of W are *not* independent even if the subsets are disjoint. This is due to the fact that $N_{W(n)}(A) = m$ directly implies $N_{W(n)}(W \setminus A) = n - m$. Thus the number of points in one subset has influence on the number of points in another subset, and hence these are not independent. This demonstrates that the binomial point process is not sufficiently appropriate as a model of CSR since the number of points is fixed. Therefore, it is necessary to consider a suitable probability distribution of the random number of points, such that the number of points in one subset cannot be predicted from the number of points in another subset.

This distribution may be derived as follows. As noted, $N_{W(n)}(A)$ follows a binomial distribution with parameters n and $p(A)$. The well-known Poisson limit theorem yields the following: if the total number of points n tends to infinity and the second parameter $p(A)$ tends to zero in such a way that the product remains fixed as $np(A) = \lambda \cdot \nu(A)$, then $N_{W(n)}(A)$ is asymptotically Poisson distributed with mean $\lambda \cdot \nu(A)$. This limit can be obtained if the region W is enlarged to fill the whole of $\mathbb{R}^d$ while n is allowed to tend to infinity. If the ratio $n/\nu(W) = \lambda$ remains fixed as n increases and W is enlarged, then the Poisson limit will hold for $N_{W(n)}(A)$ for any fixed bounded subset A of W . If there is a limiting point process N then it should have the property:

$N(A)$ is Poisson of mean $\lambda \cdot \nu(A)$ for each bounded set A .

As an implication of (2.2.5) such a limiting process should be the right model for CSR:

$N(A_1), \dots, N(A_k)$ are independent if $A_1, \dots, A_k$ are disjoint sets.

Hence the idea of assuming that the number of points in N is random and follows a Poisson distribution leads to the CSR model. The following describes a construction of this model that does not refer to limits; the limiting procedure above is used only to aid understanding.

2.2.3 The periodic binomial process

Assume that W is the rectangle with left lower vertex at the origin o and sides of lengths a and b parallel to x - and y -axis. Then the distributional properties do not change much if W is transformed to the unit square $\square$ and the points $x_i = (\xi_{i,1}, \xi_{i,2})$ become $x_i = (\xi_{i,1}, \xi_{i,2})$. Now assume that the resulting pattern in the unit square is periodically continued in all other unit squares of the plane as shown in Figure 2.1, or, more technically, consider in addition to the original points $x_i = (\xi_{i,1}, \xi_{i,2})$ infinitely many copies $x_i = (\xi_{i,1}, \xi_{i,2})$ for all integers k and l . This pattern does not change under translations which transform points with integer-valued coordinates into points with integer-valued coordinates.

It makes sense to define functional summary characteristics for the periodic binomial process that originate in the theory of stationary point processes (see Chapter 4): $D(r)$, the distribution function of the distance from an arbitrary point in

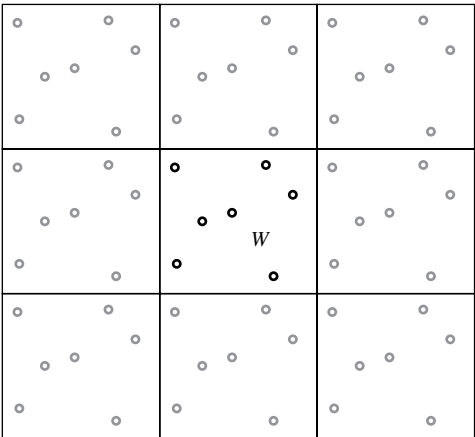


Figure 2.1 A sample of a periodic binomial process. The patterns in all squares are congruent.

the pattern to its nearest neighbour; $H_s(r)$, the distribution function of the distance from an arbitrary test point to its nearest neighbour in the pattern; and $K(r)$, the mean number of further points of the pattern in a disc of radius r centred at an arbitrary point of the pattern, divided by n . Note that the distances are measured with respect to the *torus metric* as defined on p. 184. For [1] the functions can be calculated as follows:

$$D(r) = \begin{cases} 1 - (1 - \pi r^2)^{n-1} & \text{for } r \leq \frac{1}{2}, \\ 1 - \left(1 - \sqrt{4r^2 - 1} + r^2 \left(4\arccos \frac{1}{2r} - \pi\right)\right)^{n-1} & \text{for } \frac{1}{2} < r \leq \frac{\sqrt{2}}{2}, \\ 1 & \text{for } r > \frac{\sqrt{2}}{2}; \end{cases}$$

$$H_s(r) = \begin{cases} 1 - (1 - \pi r^2)^n & \text{for } r \leq \frac{1}{2}, \\ 1 - \left(1 - \sqrt{4r^2 - 1} + r^2 \left(4\arccos \frac{1}{2r} - \pi\right)\right)^n & \text{for } \frac{1}{2} < r \leq \frac{\sqrt{2}}{2}, \\ 1 & \text{for } r > \frac{\sqrt{2}}{2}; \end{cases}$$

$$K(r) = \begin{cases} \frac{n-1}{n} \pi r^2 & \text{for } r < 1, \\ 4 + \frac{n-1}{n} \pi r^2 & \text{for } 1 \leq r < \sqrt{2}. \end{cases}$$

The proof of these formulas uses simple ideas of geometrical probability. Note that in the calculation of $D(r)$ and $K(r)$ the reference point is excluded; therefore, the exponent $n - 1$ and the factor $\frac{n-1}{n}$ appear; see the discussion at the end of Section 2.5.2.

2.2.4 Simulation of the binomial process

As explained in Section 1.9, simulation is an important method in point process statistics. This is due to the fact that for statistical inference it is often necessary to simulate from a point process model. Therefore, here the simulation of a binomial point process is described as a first simple example and as an introduction to general principles. Further examples will show that simulation approaches are often similar for more complex models and are based on the simple case considered here.

The simulation of a binomial point process is easily done by superpositioning random points independently in the required region. It is straightforward to simulate a random point that is uniformly distributed in [1]. If $\{u_n\}$ is a sequence of independent random numbers uniformly distributed in $[0, 1]$ then the points

$$x_i = (u_{2i-1}, u_{2i}) \quad \text{for } i = 1, 2, \dots, \quad (2.2.6)$$

form a sequence of independent random points uniformly distributed in [1]. Figure 2.2 shows a sample of 50 points simulated in this way.

More generally, when patterns in a d -dimensional space are considered, a sequence of random points uniformly distributed in the d -dimensional hypercube

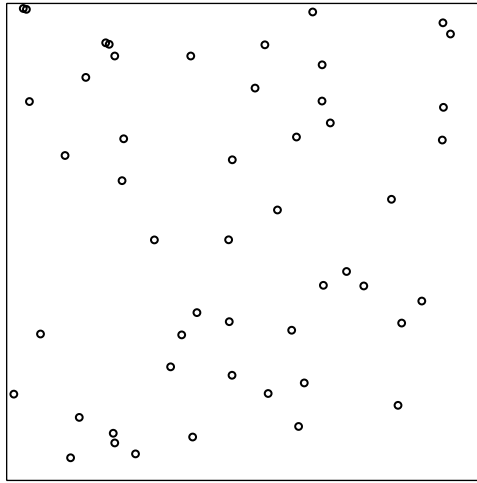


Figure 2.2 A simulation of 50 random points uniformly distributed in $[0, 1]^d$.

$[0, 1]^d$ is generated by the same mechanism, i.e.

$$x_i = (u_{(i-1)d+1}, \dots, u_{id}) \quad \text{for } i = 1, 2, \dots \quad (2.2.7)$$

Translation and scale changes may be used to produce a sequence of points uniformly distributed in any fixed rectangle or hypercube.

The simulation procedure described above is the method of choice for a large number of cases in practice, as data have been collected on a rectangle or in a cube in many applications and the binomial point process will have to be simulated in a rectangle or cube. However, in some applications observation windows with complicated shapes make the simulation approach slightly more difficult. Simulation of a uniform random point in a bounded region W of arbitrary shape is tackled using one of three main techniques, of which only the planar case is considered here; all methods may be canonically generalised to higher dimensions.

- (a) *Rejection sampling.* A rectangle R containing W is found and a sequence of independent uniform random points is simulated in R . These are rejected when they are outside W . The first point that falls in W is uniformly distributed in W . To obtain a binomial point process the whole procedure is repeated until n points have fallen in W , and these n points constitute the sample of the binomial point process. To maximise efficiency, R should be chosen to be as small as possible. Figure 2.3 illustrates the process. Note that the total number of generated points including those that were rejected is random.

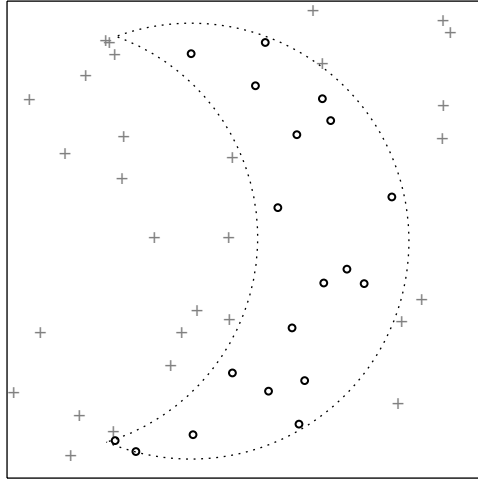


Figure 2.3 A simulated pattern of 19 random points in W based on the rectangle in Figure 2.2. The points $\circ$ are uniformly distributed in W and the points $+$ are outside W .

- (b) *Approximation.* The region W is replaced by a disjoint union of k squares (of equal or different sizes) approximating W . A random point distributed uniformly in this union is simulated by choosing a square with probability proportional to its area and then simulating a random point uniformly distributed in this square. Exact simulations for complicated regions may be obtained by combining this technique with rejection sampling (a).
- (c) *Transformation of coordinates.* If the region W exhibits some symmetry then a transformation of coordinates may be useful. For example, if W is the unit disc $b(o, 1)$, a uniform random point can be described in polar coordinates

$$x = (r, \theta) \quad \text{for } r \text{ in } [0, 1] \text{ and } \theta \in (0, 2\pi].$$

The random variables r and θ are independent; θ is uniformly distributed in $(0, 2\pi]$ and r satisfies the law

$$\mathbf{P}(r \leq t) = t^2 \quad \text{for } 0 \leq t \leq 1.$$

Thus, if u_1 and u_2 are independent random numbers uniformly distributed in $[0, 1]$ then the formulas

$$r = \sqrt{u_1} \quad \text{and} \quad \theta = 2\pi u_2$$

provide a method for simulating $x = (r, \theta)$.

Note that the binomial point process arises from the stationary Poisson point process by conditioning on n ; see property (e) in Section 2.3.2 for details. Consequently, it is not necessary to discuss specific statistical methods for assessing the hypothesis that a given point pattern is a realisation of a binomial point process; one can apply the methods for stationary Poisson point processes described in Section 2.7.

2.3 The homogeneous Poisson point process

2.3.1 Introduction

This section presents the formal definition of the homogeneous Poisson point process, or homogeneous Poisson process for short. It follows straightforwardly from the discussions in the previous section and it also essentially explains the name of the process.

A homogeneous Poisson process N is characterised by two fundamental properties which have been already mentioned as asymptotic properties in Section 2.2:

- (1) *Poisson distribution of point counts.* The number of points of N in any bounded set B follows a Poisson distribution with mean $\lambda \cdot \nu(B)$ for some constant λ ; readers unfamiliar with the definition of the Poisson distribution may refer to formula (2.3.2) and p. 106.
- (2) *Independent scattering.* The numbers of points of N in k disjoint sets form k independent random variables, for arbitrary k .

Property (2) is also known as the ‘completely random’ or ‘purely random’ property. Note that this property does not hold for the centres of hard objects that are randomly distributed in space. Models have been developed which may be suitable for these patterns but they are beyond the scope of the Poisson process theory; see Section 6.5.4 (the random sequential adsorption (RSA) model) and 6.6.1 (Gibbs hard-core process).

The number λ in (1) is the characteristic parameter, called the *intensity* or *point density*, of the homogeneous Poisson process. It describes the mean number of points to be found in a unit volume and is given by

$$\lambda \cdot \nu(B) = \mathbf{E}(N(B)) \quad \text{for all bounded sets } B. \quad (2.3.1)$$

In the following, λ is always positive and finite. (If $\lambda = 0$, the point pattern contains no points and an infinite λ corresponds to a pathological case.)

2.3.2 Basic properties

Let N be a homogeneous Poisson process with intensity λ . Once the intensity λ is known, the whole distribution of the homogeneous Poisson process can be determined from properties (1) and (2).

(a) *One-dimensional distributions.* Property (1) implies that

$$\mathbf{P}(N(B) = n) = \frac{\lambda^n (\nu(B))^n}{n!} \exp(-\lambda \nu(B)) \quad \text{for } n = 0, 1, \dots \quad (2.3.2)$$

This means that $N(B)$ has a Poisson distribution with parameter $\lambda \nu(B)$.

(b) *Finite-dimensional distributions.* Properties (1) and (2) imply that if $B_1, \dots, B_k$ are disjoint bounded sets then $N(B_1), \dots, N(B_k)$ are independent Poisson random variables with means $\lambda \nu(B_1), \dots, \lambda \nu(B_k)$. Thus

$$\begin{aligned} \mathbf{P}(N(B_1) = n_1, \dots, N(B_k) = n_k) \\ = \frac{\lambda^{n_1 + \dots + n_k} (\nu(B_1))^{n_1} \dots (\nu(B_k))^{n_k}}{n_1! \dots n_k!} \exp\left(-\sum_{i=1}^k \lambda \nu(B_i)\right). \end{aligned} \quad (2.3.3)$$

From this formula the joint probabilities $\mathbf{P}(N(B_1) = n_1, \dots, N(B_k) = n_k)$ may be evaluated for general (possibly overlapping) sets $B_1, \dots, B_k$.

(c) *Stationarity and isotropy.* A point process $N = \{x_n\}$ is stationary if the translated process $N_x = \{x_n + x\}$ has the same distribution for all x in $\mathbb{R}^d$ (recall Section 1.6.3). A point process is isotropic if its distribution is invariant with respect to rotations, i.e. N and $\mathbf{r}N = \{\mathbf{r}x_n\}$ have the same distribution for every rotation $\mathbf{r}$ about the origin. A process is motion-invariant if it possesses both these properties. The homogeneous Poisson process N is defined by properties (1) and (2) above and the specification of the intensity λ . These properties and the characteristic λ are clearly invariant under rotation and translation. Therefore the homogeneous Poisson process N has to be stationary and isotropic, that is to say, motion-invariant. Typically, the older, more traditional name ‘homogeneous’ Poisson process is used rather than the term ‘stationary’ Poisson process.

That the stationarity and isotropy properties do indeed hold may be verified directly by establishing that the finite-dimensional distributions above remain the same no matter whether one uses a homogeneous Poisson process N or its translation N_x or its rotation $\mathbf{r}N$.

The Poisson process has a further ‘conservation property’ with respect to linear transformations, where the fundamental properties are retained but only the intensity changes. This property has a quite natural interpretation and application. Consider a pattern of trees observed as aerial images, where the images have been taken at different heights and angles. This results in different patterns, which are linear transformations of the same original pattern. If the true pattern is Poisson, then the same is true for all transformed patterns, but the intensities (mean numbers of trees

per unit area on the images) do of course differ. The difference in intensity can easily be calculated such that the intensity of the original pattern can be determined.

More technically, the property is formulated as follows. Let $\mathbf{A}$ be a non-singular linear mapping from $\mathbb{R}^d$ to $\mathbb{R}^d$. If N is a homogeneous Poisson process with intensity λ , then $\mathbf{A}N = \{\mathbf{A}x : x \in N\}$ is also a homogeneous Poisson process and its intensity is $\lambda \cdot \det(\mathbf{A}^{-1})$, where $\det(\mathbf{A}^{-1})$ is the determinant of the inverse of $\mathbf{A}$. A simple special case is the linear transformation

$$x = (\xi_1, \xi_2) \rightarrow \left(\frac{\xi_1}{a}, \frac{\xi_2}{b} \right),$$

which transforms a rectangle of side lengths a and b into the unit square. Clearly, the intensity of the transformed process is $ab\lambda$.

(d) *Void probabilities.* The *void probabilities* v_K of a point process are the probabilities of there being no point of the process in given test sets K :

$$v_K = \mathbf{P}(N(K) = 0).$$

If N is a homogeneous Poisson process,

$$v_K = \exp(-\lambda \nu(K)). \quad (2.3.4)$$

The contact distribution functions are closely related to the void probabilities. An important special case is the *spherical contact distribution function* as introduced in (1.5.6),

$$H_s(r) = 1 - \mathbf{P}(N(b(o, r)) = 0). \quad (2.3.5)$$

By (2.3.4) this yields for the Poisson process case

$$H_s(r) = 1 - \exp(-\lambda b_d r^d) \quad \text{for } r \geq 0. \quad (2.3.6)$$

This is the distribution function of the distance from o to the nearest point of N ; see Section 4.2.5. The mean and variance of this distribution for $d = 2$ are given by

$$m_D = \frac{1}{2\sqrt{\lambda}} \quad (2.3.7)$$

and

$$\sigma_D^2 = \frac{1}{\pi\lambda} - \frac{1}{4\lambda}. \quad (2.3.8)$$

The idea can be generalised by considering an arbitrary set B with $\nu(B) > 0$ and $o \in B$. The *contact distribution function* $H_B(r)$ (with respect to B) is given by

$$H_B(r) = 1 - v_{rB} = 1 - \mathbf{P}(N(rB) = 0) \quad \text{for } r \geq 0.$$

It makes sense to consider, for example, the case of a rectangular B . The spherical contact d.f. is $H_B(r)$ with $B = b(o, 1)$.

(e) *Conditioning and binomial point processes.* Consider the restriction of the homogeneous Poisson process N to a bounded set W conditional on $N(W) = n$. This ‘conditioning’ yields a new point process which is actually the binomial point process in W with n points.

This assertion can be proved by showing that the finite-dimensional distributions of the two processes coincide. In fact, it suffices to consider only the void probabilities. If K is a bounded subset of W then the void probability for K of the conditioned homogeneous Poisson process is given by

$$\begin{aligned} \mathbf{P}(N(K) = 0 | N(W) = n) &= \frac{\mathbf{P}(N(K) = 0, N(W) = n)}{\mathbf{P}(N(W) = n)} \\ &= \frac{\mathbf{P}(N(K) = 0) \mathbf{P}(N(W \setminus K) = n)}{\mathbf{P}(N(W) = n)} = \frac{(\nu(W) - \nu(K))^n}{\nu(W)^n} \end{aligned}$$

by (a) after substitution and cancellation. This formula coincides with (2.2.4) for the void probabilities of the binomial point process.

2.3.3 Characterisations of the homogeneous Poisson process

One may ask whether properties (1) and (2) are both necessary to completely characterise a homogeneous Poisson process? Indeed, the properties are not logically independent; see Kingman (1993). Rényi (1967) showed that property (1) implies property (2); the Poisson distribution property forces the point process to have the independent scattering property as described in (e) above. One may thus suspect that property (2) is superfluous.

However, if property (1) is weakened, property (2) may be necessary for the characterisation of the homogeneous Poisson process. More specifically, property (2) does *not* follow from (1) if (1) holds only for the class of all connected subsets of $\mathbb{R}^d$. Moreover, Moran (1976) shows that it is not enough to assume the independence and Poisson distribution of counts of points in k arbitrary disjoint convex sets for some fixed k . In particular, one can construct point processes with Poisson counts in one size of quadrats which are nevertheless not Poisson processes (Dale, 1999).

Another set of properties also characterises the Poisson process, and should be borne in mind as it frequently provides a *prima facie* case for assuming that an empirical point pattern is a realisation of a homogeneous Poisson process. This characterisation asserts that a process must be a homogeneous Poisson process if the following three properties are satisfied:

- (i) *Simplicity.* Two points never coincide, i.e. the process is a *simple* point process; see Section 1.5.
- (ii) *Stationarity.* This has been defined above; see also Section 1.6.

- (iii) *Independent scattering*. This is property (2) as given at the beginning of this section.

The most important condition is that of independent scattering. In effect, it asserts that there is no interaction between the points of the pattern, and it may be tested statistically. In some cases, independent scattering is suggested by underlying biological or physical theories. Whether this is plausible or not, it frequently provides the starting point for a statistical analysis, even if only as a null hypothesis.

Note that a Poisson process is sometimes used as a ‘white noise’ process, corresponding to the independent and identically distributed (i.i.d.) random variables in a times series context, which model completely random disturbance effects. A good explanation of the fact that point patterns often behave like samples of Poisson processes is the *Poisson convergence theorem* which says that the superposition of many thin point processes is a Poisson process (see Kallenberg, 2002, Theorem 16.18).

2.4 Simulation of a homogeneous Poisson process

As the above sections have shown, many formulas have been derived for the Poisson process and many important characteristics can therefore be calculated exactly. Nevertheless, it is still important to be able to also simulate this process. There are three main reasons for this:

- There are situations where even Poisson process characteristics are difficult to obtain, for example in the context of tests, as shown in Section 2.7.7 below.
- The Poisson process is a building block for more complex processes. These can often only be explored by simulation. The simulation of many more complex models is based on the simulation of a Poisson process.
- To motivate the use of simulations in point process statistics and to facilitate understanding, it might be helpful initially to use a simple example.

The starting point for the simulation of a homogeneous Poisson process in a bounded set W is property (e) in Section 2.3.2 that conditioning on the total number of points in W yields a binomial point process. As a result, once the total number of points has been determined, simulation methods for the binomial process as described in Section 2.2 may be applied.

Thus the simulation essentially consists of two steps. First, the number of points in W is determined by simulating a Poisson random variable, and then the positions of the points in W are determined by simulating a binomial point process in W with the number of points determined in the first step.

The following describes two possibilities for the first step, the simulation of a Poisson random variable. Which of the two methods is more appropriate depends on the mean $\lambda \cdot \nu(W)$ one intends the simulated pattern to have.

If $\lambda \cdot \nu(W)$ is small, a linear (or one-dimensional) Poisson process may be simulated, exploiting the fact that for such a process the (one-dimensional) inter-point distances are independent exponential random variables. For this purpose random numbers e_i from an exponential distribution with mean 1 are generated. This can be achieved by applying the inversion method to uniform random numbers. If u_i is uniform on $[0, 1]$ then $e_i = -\ln(u_i)$ is as required. The desired Poisson random variable is the smallest n for which

$$\sum_{i=1}^{n+1} e_i > \lambda \cdot \nu(W). \quad (2.4.1)$$

Since addition of logarithms is equivalent to multiplication, the Poisson variable can be determined more elegantly by

$$\prod_{i=1}^{n+1} u_i < \exp(-\lambda \cdot \nu(W)). \quad (2.4.2)$$

If $\lambda \cdot \nu(W)$ is large, some form of rejection technique can be used; see Devroye (1986) and Ripley (1987). Alternatively, the central limit theorem may simply be exploited. It states that for large μ , a Poisson random variable with mean μ approximately follows a Gaussian distribution with mean μ and variance μ . This is because the mean and variance coincide for a Poisson distribution. Thus a Gaussian random number may be generated by well-known methods and then rounded to an integer.

In the special case where W is a disc or a sphere, another method may be used based on polar coordinates, as described on p. 65. This method, developed by Quine and Watson (1984), is more elegant than the rejection method and is called ‘radial generation’.

2.5 Model characteristics

2.5.1 Moments and moment measures*

Just as moments are important characteristics for ‘standard’ random variables, so are the corresponding entities for the Poisson process and for general point processes. Section 1.5 shows that the moments of a point process are given by measures. These *moment measures* are described here for the specific case of the homogeneous Poisson process, where explicit formulas are known. These may appear rather technical to the applied reader, who might prefer to skip this section. At a later stage, however, when it has become clearer that the theory is relevant, it might be useful to return to this section.

Consider the homogeneous Poisson process N and take a bounded subset B of $\mathbb{R}^d$. Then $N(B)$ is a random variable with first moment or mean

$$\lambda \nu(B) = \mathbf{E}(N(B)). \quad (2.5.1)$$

If B_1 and B_2 are two bounded sets, $N(B_1)$ and $N(B_2)$ are two random variables with a non-centred covariance

$$\mu^{(2)}(B_1 \times B_2) = \mathbf{E}(N(B_1)N(B_2)).$$

The second-order quantity $\mu^{(2)}(B_1 \times B_2)$ is evaluated by using properties (1) and (2) in Section 2.3.1 as follows. Note that both B_1 and B_2 may be decomposed into disjoint unions

$$\begin{aligned} B_1 &= (B_1 \cap B_2) \cup (B_1 \setminus B_2), \\ B_2 &= (B_1 \cap B_2) \cup (B_2 \setminus B_1). \end{aligned}$$

Applying property (2) and the fact that N satisfies

$$N(A \cup B) = N(A) + N(B)$$

for arbitrary non-intersecting A and B , one can establish

$$\begin{aligned} \mu^{(2)}(B_1 \times B_2) &= \mathbf{E}(N(B_1)N(B_2)) \\ &= \mathbf{E}(N(B_1 \setminus B_2)) \cdot \mathbf{E}(N(B_2 \setminus B_1)) \\ &\quad + \mathbf{E}(N(B_1 \cap B_2)) \cdot \mathbf{E}(N(B_2 \setminus B_1)) \\ &\quad + \mathbf{E}(N(B_1 \setminus B_2)) \cdot \mathbf{E}(N(B_1 \cap B_2)) + \mathbf{E}((N(B_1 \cap B_2))^2) \\ &= \mathbf{E}(N(B_1)) \cdot \mathbf{E}(N(B_2)) + \mathbf{E}((N(B_1 \cap B_2))^2) \\ &\quad - (\mathbf{E}(N(B_1 \cap B_2)))^2. \end{aligned}$$

Property (1) ensures that $N(B_1 \cap B_2)$ is Poisson with mean and variance $\mathbf{E}(N(B_1 \cap B_2))$. Using (2.5.1) the final formula can be derived:

$$\begin{aligned} \mu^{(2)}(B_1 \times B_2) &= \mathbf{E}(N(B_1))\mathbf{E}(N(B_2)) + \mathbf{E}(N(B_1 \cap B_2)) \\ &= \lambda^2 \cdot \nu(B_1) \cdot \nu(B_2) + \lambda \cdot \nu(B_1 \cap B_2). \end{aligned} \quad (2.5.2)$$

Thus, the *second-order moment measure* $\mu^{(2)}$ can be expressed in terms of λ and the volume measure ν . Note that $\mu^{(2)}$ is a measure on $\mathbb{R}^d \times \mathbb{R}^d$.

Variances and covariances can be calculated directly from the second-order moment measure (this is also true for general point processes). The relevant formulas are

$$\begin{aligned}\text{var}(N(B)) &= \mu^{(2)}(B \times B) - (\mathbf{E}(N(B)))^2, \\ \text{cov}(N(B_1), N(B_2)) &= \mu^{(2)}(B_1 \times B_2) - \mathbf{E}(N(B_1)) \cdot \mathbf{E}(N(B_2))\end{aligned}$$

for all sets B , B_1 and B_2 . These equations follow immediately from the definitions of variance and covariance. In the case of the homogeneous Poisson process one can use (2.5.2) or calculate directly from properties (1) and (2) to show that

$$\text{cov}(N(B_1), N(B_2)) = \lambda \cdot \nu(B_1 \cap B_2). \quad (2.5.3)$$

Note that $\mu^{(2)}(B_1 \times B_2)$ can also be expressed as the expectation of this sum:

$$\begin{aligned}\mu^{(2)}(B_1 \times B_2) &= \mathbf{E}(\#\{(x_1, x_2) : x_1 \in N \cap B_1, x_2 \in N \cap B_2\}) \\ &= \mathbf{E}\left(\sum_{x_1, x_2 \in N} \mathbf{1}_{B_1}(x_1) \mathbf{1}_{B_2}(x_2)\right).\end{aligned}$$

That is, $\mu^{(2)}(B_1 \times B_2)$ is the mean number of pairs of points (x_1, x_2) in N with $x_1 \in B_1$ and $x_2 \in B_2$.

The two terms in (2.5.2) correspond to the dissection of this sum into the sum over distinct pairs of points $x_1, x_2 \in N$ and the sum over equal points $x_1 = x_2 \in N$. For some purposes it is convenient to subtract out the second of these terms. The result is the *second-order factorial moment measure* $\alpha^{(2)}$ given by

$$\begin{aligned}\alpha^{(2)}(B_1 \times B_2) &= \mathbf{E}(\#\{(x_1, x_2) : x_1 \in N \cap B_1, x_2 \in N \cap B_2, x_1 \neq x_2\}) \\ &= \mathbf{E}\left(\sum_{x_1, x_2 \in N}^{\neq} \mathbf{1}_{B_1}(x_1) \mathbf{1}_{B_2}(x_2)\right).\end{aligned}$$

Here $\sum^{\neq}$ denotes the summation over all pairs (x_1, x_2) such that $x_1 \neq x_2$.

Since the terms $\mu^{(2)}(B_1 \times B_2)$ and $\alpha^{(2)}(B_1 \times B_2)$ differ only by the expectation of the sum

$$\sum_{x_1, x_2 \in N: x_1 = x_2} \mathbf{1}_{B_1}(x_1) \mathbf{1}_{B_2}(x_2) = \sum_{x \in N} \mathbf{1}_{B_1 \cap B_2}(x),$$

we have

$$\mu^{(2)}(B_1 \times B_2) = \alpha^{(2)}(B_1 \times B_2) + \Lambda(B_1 \cap B_2).$$

In the case of the homogeneous Poisson process,

$$\alpha^{(2)}(B_1 \times B_2) = \lambda^2 \cdot \nu(B_1) \cdot \nu(B_2) = \mathbf{E}(N(B_1)) \cdot \mathbf{E}(N(B_2)). \quad (2.5.4)$$

Thus the *second-order product density* $\varrho^{(2)}$ has the very simple form

$$\varrho^{(2)}(x_1, x_2) = \lambda^2 \quad \text{for } x_1, x_2 \text{ in } \mathbb{R}^d. \quad (2.5.5)$$

By analogy with higher-order moments for random variables, higher-order moment measures may be defined for point processes. The *nth-order moment measure* $\mu^{(n)}$ is a measure on $\mathbb{R}^{nd}$ defined by

$$\begin{aligned} \mu^{(n)}(B_1 \times \cdots \times B_n) &= \mathbf{E}(N(B_1) \cdot \cdots \cdot N(B_n)) \\ &= \mathbf{E}\left(\sum_{x_1, \dots, x_n \in N} \mathbf{1}_{B_1}(x_1) \cdot \cdots \cdot \mathbf{1}_{B_n}(x_n)\right) \end{aligned}$$

for sets $B_1, \dots, B_n$. The *nth-order factorial moment measure* $\alpha^{(n)}$ is a measure on $\mathbb{R}^{nd}$ given by

$$\alpha^{(n)}(B_1 \times \cdots \times B_n) = \mathbf{E}\left(\sum_{x_1, \dots, x_n \in N}^{\neq} \mathbf{1}_{B_1}(x_1) \cdot \cdots \cdot \mathbf{1}_{B_n}(x_n)\right),$$

where $\sum^{\neq}$ denotes summation over n -tuples $(x_1, \dots, x_n)$ of distinct points. The corresponding *nth-order product density* $\varrho^{(n)}$ is given by the simple formula

$$\varrho^{(n)}(x_1, \dots, x_n) = \lambda^n \quad (2.5.6)$$

and $\alpha^{(n)}$ has the form

$$\alpha^{(n)}(B_1 \times \cdots \times B_n) = \lambda^n \cdot \nu(B_1) \cdot \cdots \cdot \nu(B_n). \quad (2.5.7)$$

2.5.2 The Palm distribution of a homogeneous Poisson process

In point process theory the so-called ‘typical’ point of a point process N is frequently considered. Informally, this is a point that has been chosen by a selection procedure in which every point of the process has the same chance of being selected. For example, the nearest-neighbour distance distribution function $D(r)$ describes the distribution of the distance from a typical point x in N to the nearest point in $N \setminus \{x\}$, i.e. to the nearest neighbour of x in N . Mathematically, the idea of the typical point is precisely described by means of the Palm distribution theory, as sketched in Section 4.1. In intuitive terms, the Palm distribution probabilities are

conditional probabilities of point process events given that a point (the typical point) has been observed at a specific location.

There are two heuristic approaches to the definition of the Palm distribution. The global approach is described in Section 4.1, while the local approach is used here. It is demonstrated in the following example.

Calculation of the nearest neighbour distance distribution function for the homogeneous Poisson process

Consider a conditional nearest-neighbour distance d.f. $D_\varepsilon(r)$, i.e. assume that there is a point of N in $b(o, \varepsilon)$. The conditional probability

$$D_\varepsilon(r) = 1 - \mathbf{P}(N(b(o, r) \setminus b(o, \varepsilon)) = 0 \mid N(b(o, \varepsilon)) = 1)$$

is the probability that the distance from a point in the small sphere $b(o, \varepsilon)$ to its nearest neighbour in N is smaller than r , under the condition that there is indeed a point of N in the small sphere. It is well defined for positive ε smaller than r , since

$$\mathbf{P}(N(b(o, \varepsilon)) = 1) = \lambda b_d \varepsilon^d \exp(-\lambda b_d \varepsilon^d)$$

is positive. Using the definition of conditional probability and property (2) of the homogeneous Poisson process,

$$\begin{aligned} D_\varepsilon(r) &= 1 - \frac{\mathbf{P}(N(b(o, r) \setminus b(o, \varepsilon)) = 0) \mathbf{P}(N(b(o, \varepsilon)) = 1)}{\mathbf{P}(N(b(o, \varepsilon)) = 1)} \\ &= 1 - \mathbf{P}(N(b(o, r) \setminus b(o, \varepsilon)) = 0) \\ &= 1 - \exp(-\lambda(\nu(b(o, r)) - \nu(b(o, \varepsilon)))). \end{aligned}$$

It is reasonable to regard the nearest-neighbour distance d.f. $D(r)$ as the limit of the above as $\varepsilon \rightarrow 0$. Setting $D(r) = \lim_{\varepsilon \rightarrow 0} D_\varepsilon(r)$ yields the result

$$D(r) = 1 - \exp(-\lambda \nu(b(o, r)))$$

or

$$D(r) = 1 - \exp(-\lambda b_d r^d) \quad \text{for } r \geq 0. \quad (2.5.8)$$

This result can be established in a rigorous way by means of the global approach; see Stoyan et al. (1995).

The right-hand sides of (2.5.8) and (2.3.6) are equal, i.e. the spherical contact d.f. and the nearest-neighbour distance d.f. of the homogeneous Poisson process coincide,

$$D(r) = H_s(r) \quad \text{for } r \geq 0. \quad (2.5.9)$$

The mean and variance corresponding to $D(r)$ in the case $d = 2$ are thus given by (2.3.7) and (2.3.8).

The distribution functions $D_2(r)$, $D_3(r)$, $\dots$ of the distances to the 2nd, 3rd, $\dots$ nearest neighbours are

$$D_k(r) = 1 - \sum_{j=0}^{k-1} \exp(-\lambda \pi r^2) \frac{(\lambda \pi r^2)^j}{j!} \quad \text{for } r \geq 0, \quad (2.5.10)$$

and the corresponding probability density functions are

$$d_k(r) = \frac{2(\lambda \pi r^2)^k}{r(k-1)!} \exp(-\lambda \pi r^2) \quad \text{for } r \geq 0;$$

see Figure 2.4.

The corresponding j th moments are

$$m_{k,j} = \frac{\Gamma(k + \frac{1}{2}j)}{(k-1)!(\lambda \pi)^{j/2}} \quad \text{for } j = 1, 2, \dots,$$

and the position of the mode (maximum of density function) is

$$r_k = \sqrt{\frac{k - \frac{1}{2}}{\lambda \pi}}.$$

Similar to the case of $D(r)$ above, the *reduced second-order moment measure* function or Ripley's K -function $K(r)$ can also be calculated explicitly for the homogeneous Poisson process. The term $\lambda K(r)$ is the mean number of points other

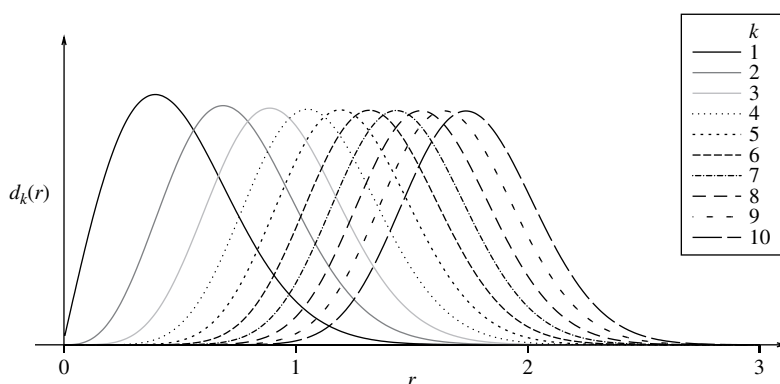


Figure 2.4 Density functions of the distances to the k th nearest neighbours for a homogenous Poisson process of intensity 1, for $k = 1, 2, \dots, 10$.

than the typical point in a ball of radius r centred at the typical point. In the Poisson process case

$$\lambda K(r) = \mathbf{E}(N(b(o, r))), \quad (2.5.11)$$

i.e.

$$\lambda K(r) = \lambda b_d r^d$$

and so

$$K(r) = b_d r^d \quad \text{for } r \geq 0. \quad (2.5.12)$$

The so-called L -function is obtained by

$$L(r) = \left(\frac{K(r)}{b_d} \right)^{\frac{1}{d}}$$

as

$$L(r) = r \quad \text{for } r \geq 0, \quad (2.5.13)$$

and, similarly, the pair correlation function $g(r)$ is given by

$$g(r) = 1 \quad \text{for } r \geq 0, \quad (2.5.14)$$

due to the general relation to $K(r)$,

$$g(r) = \frac{dK(r)}{dr} \bigg/ db_d.$$

Using the symbol $\mathbf{P}_o$ for the Palm distribution, formulas (2.5.9) and (2.5.12) can be rewritten as

$$\begin{aligned} & \mathbf{P}(\text{distance from typical point to nearest neighbour in } N \leq r) \\ &= \mathbf{P}_o(\text{distance from } o \text{ to nearest neighbour in } N \leq r) \\ &= \mathbf{P}(\text{distance from } o \text{ to nearest neighbour in } N \leq r) \end{aligned}$$

and

$$\begin{aligned} & \mathbf{E}(\text{number of points in sphere of radius } r \text{ around typical point,} \\ & \text{which is not counted}) \\ &= \mathbf{E}_o(N(b(o, r) \setminus \{o\})) = \mathbf{E}(N(b(o, r))). \end{aligned}$$

This means that Palm characteristics of homogenous Poisson processes can be calculated as stationary characteristics, if a point in the origin o is added to N . This is the important *Slivnyak–Mecke theorem*, which states that the Palm distribution of a homogeneous Poisson process coincides with that of the point process obtained by adding the origin o to the homogeneous Poisson process. In particular,

$$\mathbf{P}_o(N \in \mathcal{A}) = \mathbf{P}(N \cup \{o\} \in \mathcal{A})$$

and

$$\mathbf{E}_o(N(B)) = \mathbf{E}(N(B)) + \mathbf{1}_B(o).$$

As in Section 1.6, $\mathcal{A}$ denotes a point process event; for example, ‘ $N \in \mathcal{A}$ ’ may mean that the point process N has n points in some set B .

The above theorem will be applied repeatedly in this book. The interested reader may find a formal proof in Stoyan et al. (1995). Heuristically, the theorem says that the probabilistic behaviour of a homogeneous Poisson process is the same whether it is seen from an independent test location or from a randomly chosen point in the process.

A comparison of the formulas for $D(r)$ and $K(r)$ for the Poisson process and the periodic binomial process reveals that these processes differ slightly. This shows again that they do not model the same concept and hence that the periodic binomial process is not an appropriate CSR model.

2.5.3 Summary characteristics of the homogeneous Poisson process

This section provides an overview of the results for the characteristics of the homogeneous Poisson process (see Tables 2.1–2.4). In practice, when a pattern’s

Table 2.1 Functional summary characteristics for the homogeneous Poisson process.

Characteristic	Formula	Page
K -function, $K(r)$	$b_d r^d$	77
L -function, $L(r)$	r	77
Pair correlation function, $g(r)$	1	77
Nearest-neighbour distance d.f., $D(r)$	$1 - \exp(-\lambda b_d r^d)$	75
Spherical contact d.f., $H_s(r)$	$1 - \exp(-\lambda b_d r^d)$	68
J -function, $J(r)$	1	91

λ = intensity = point density, r = inter-point distance, $b_d = 2$ ($d = 1$), π ($d = 2$) and $\frac{4}{3}\pi$ ($d = 3$).

Table 2.2 Numerical summary characteristics for the homogeneous Poisson process.

Characteristic	Numerical value	Page
Index of dispersion, ID	1	195
Aggregation index, CE	1	196
Pielou index of randomness, PI	1	196
Mean-direction index, $\bar{R}_4$	1.799	93

Table 2.3 Formulas for the Poisson–Voronoi tessellation, planar case.

Characteristic	Mean	Variance
Cell area	λ^{-1}	$0.2802\lambda^{-2}$
Perimeter	$4\sqrt{\lambda}/\lambda$	$0.9455\lambda^{-1}$
Number of vertices	6	1.7808

Table 2.4 Formulas for the Poisson–Voronoi tessellation, spatial case.

Characteristic	Mean	Variance
Cell volume	λ^{-1}	$0.179\lambda^{-2}$
Surface	$5.821\lambda^{-2/3}$	$2.19\lambda^{-4/3}$
Number of vertices	27.071	43.99

properties are assessed with regard to CSR, these formulas are regularly applied. They will also be used as a reference throughout this book when other processes are compared to the homogeneous Poisson process.

Formulas for further tessellation characteristics may be found in Stoyan et al. (1995). The exact distributions of area, volume, perimeter, surface, and edge number are unknown. For volume and area, approximations have been derived which use gamma distributions.

2.6 Estimating the intensity

The most fundamental statistical question for the homogeneous Poisson process concerns the estimation of the intensity λ , which is the only parameter in the model. Several estimation methods are available. In a specific situation, the sampling conditions determine which estimation method may be the most suitable.

Counting method

If all points in the sampling window W can be counted, the number of points $N(W)$ divided by the area or volume $\nu(W)$ of the window is the natural estimator:

$$\hat{\lambda} = \frac{N(W)}{\nu(W)}. \quad (2.6.1)$$

This estimator has some nice statistical properties as it is unbiased and the maximum likelihood estimator. Its variance is

$$\text{var}(\hat{\lambda}) = \frac{\lambda}{\nu(W)}. \quad (2.6.2)$$

Note that $\hat{\lambda}$ is also unbiased for *all* stationary point processes, even for those that are not Poisson processes; see Section 4.2.3.

Distance methods

In applications, it may be practically impossible (e.g. if it is too time-consuming) to count all points. In this case, distance methods can be used. The idea behind these methods is to measure the distances to points from deterministic test locations or from points in the process. Unfortunately, this old idea (see Section 1.3.1) can only be applied with acceptable precision if the distribution of the point process is known. As noted, the distribution is not explicitly known for most point processes, and so the only case where it may be realistically applied is the Poisson process.

Distance methods for the Poisson case are discussed in detail in Diggle (2003) and Krebs (1998). Here, only the case of deterministic test locations is considered, assuming that the distances between the locations are large enough for the nearest-neighbour distances from the test locations to be approximately independent.

Point-quarter method

More information can be obtained from each test location (indexed by j) by determining four nearest-neighbour distances, namely those in the north-east quadrant,

$h_{1j}, \dots$, and in the north-west quadrant, h_{4j} . Under the assumption of a Poisson process, for each of the test locations the nearest-neighbour distance in a quadrant follows the distribution function

$$H_{s/4}(r) = 1 - \exp\left(-\frac{\pi}{4}\lambda r^2\right) \quad \text{for } r \geq 0.$$

If the quadruplets of nearest-neighbour distances for different locations were independent (which is not exactly true but holds approximately if the inter-test-point distances are large) λ can be estimated by the maximum likelihood method (Cottam et al., 1953; Diggle, 1982):

$$\hat{\lambda} = \frac{4}{\pi \bar{h}^2} \quad (2.6.3)$$

with

$$\bar{h}^2 = \sum_{j=1}^k \sum_{i=1}^4 h_{ij}^2 / 4k,$$

where k is the number of test locations.

Confidence intervals

Confidence intervals for λ can be based on (2.6.1) since $\hat{\lambda} \cdot \nu(W) = N(W)$ has a Poisson distribution. For large $N(W)$, Armitage et al. (2001) and Krebs (1998) provide a simple approximate $100(1 - \alpha)\%$ confidence interval for λ , employing the normal approximation and a continuity correction. For example,

$$\left(\frac{z_{\alpha/2}}{2} - \sqrt{N(W)}\right)^2 \leq \lambda \nu(W) \leq \left(\frac{z_{\alpha/2}}{2} + \sqrt{N(W) + 1}\right)^2. \quad (2.6.4)$$

Here the $z_{\alpha/2}$ are quantiles of the standard normal distribution:

$$z_{\alpha/2} = 1.65, 1.96, 2.58, \quad \text{for } \alpha = 0.10, 0.05, 0.01.$$

In practice, this confidence interval is a simple tool for determining the window size required for a given accuracy of estimation. If δ is the desired (full) width of the confidence interval and α is the required confidence level, then

$$\delta \cdot \nu(W) \simeq \left(\frac{z_{\alpha/2}}{2} + \sqrt{\lambda \nu(W) + 1}\right)^2 - \left(\frac{z_{\alpha/2}}{2} - \sqrt{\lambda \nu(W)}\right)^2$$

yields the approximation

$$\nu(W) \simeq \frac{4z_{\alpha/2}^2 \lambda}{\delta^2}, \quad (2.6.5)$$

where λ itself must be estimated based on data which perhaps have been derived from a pilot study or by using a priori information.

Example 2.1. Gold particles: intensity estimation

Consider the pattern of gold particles discussed on p. 6 and assume for the moment that it can be modelled by a homogeneous Poisson process. Formula (2.6.4) yields the confidence interval (0.000 754, 0.000 984) for λ for $\alpha = 0.05$. For the same α and $\lambda = 8 \cdot 10^{-4}$ and $\delta = 10^{-4}$, Formula (2.6.5) suggests a window size of 1 229 300. That is to say, a square of size 1100×1100 is sufficient to obtain 95 % confidence with an accuracy of 10^{-4} . If the true point process is not Poisson, this sample size may still be used as an approximation, which is probably a little too large.

Testing homogeneity of point density

Often, the observation window is large and the question may arise whether the point distribution is really homogeneous (i.e. whether the intensity function $\lambda(x)$ is really constant). For this purpose, the sampling window W may be split into two subregions W_1 and W_2 . To test the hypothesis that the point densities in W_1 and W_2 are equal, one may consider the observed numbers of points n_1 and n_2 and subregion measures $\nu(W_1)$ and $\nu(W_2)$.

Under the null hypothesis of equality, i.e. the hypothesis that the whole pattern is a realisation a homogeneous Poisson process, the quantity

$$F = \frac{\nu(W_1)(2n_2 + 1)}{\nu(W_2)(2n_1 + 1)},$$

follows approximately the F -distribution with $2n_1 + 1$ and $2n_2 + 1$ degrees of freedom. Note that this assumes that the order of the subscripts 1 and 2 is chosen such that F is greater than one. The equality hypothesis is rejected at a significance level α if

$$F > F_{2n_1+1, 2n_2+1; \alpha/2}.$$

Formally this test is correct only if the subregions W_1 and W_2 are chosen a priori, before collecting the data, perhaps to investigate specific possibilities of a trend in the point density. For example, such a trend might relate the point density of tree locations to some measure of nutrient levels or the water contents in the soil. It is not appropriate to apply the test a posteriori to a given point pattern that has unexpected subregions of very high or very low point densities on inspection.

2.7 Testing complete spatial randomness

2.7.1 Introduction

Testing the CSR hypothesis is an important part of exploratory data analysis of point patterns: if the hypothesis is accepted, one can assume that the given point pattern is completely spatially random, which has two main consequences:

- (a) There is no need to consider a more complicated model; the simple Poisson process model can be used, with all its consequences.
- (b) If no additional information or data on underlying processes is available it is not possible to find indicators of interesting interaction between the points based on the geometry of the observed pattern alone.

Using tests of CSR, Tomppo (1986) showed that 30 % of the permanent inventory plots in Finland of thinned forests in mineral soil areas can be considered CSR. But note with respect to point (b) that there are examples from ecology which show that Poisson-process-like point patterns can result from ecological processes operating on clustered patterns. Note, further, that real patterns are never truly Poisson process patterns; not rejecting the CSR hypothesis only means that the pattern is ‘close to’ a Poisson process sample.

If the CSR hypothesis is rejected, the more interesting part of point process statistics begins, in particular the search for spatial correlations in the given pattern. Nevertheless, the initial analysis provides valuable information on the direction of the deviation from CSR, as well as on the cause of the deviation to guide further analysis.

A large number of tests of the CSR hypothesis have been developed, and research in this area is still ongoing. Many of the summary characteristics of point processes can be used to construct such tests, and those that yield extremal or simple values in the Poisson case are particularly successful. Experience shows that it is not possible to derive a ‘best’ test if it is only based on a single criterion, and any test is only capable of assessing particular aspects of CSR behaviour. Which test is the most appropriate in a given situation depends on the nature of the alternative hypothesis envisaged. In practical applications, the choice of the appropriate test also depends on the limitations imposed by sampling methods. Nevertheless, rejection of the CSR hypothesis by any of the tests means final rejection. (Perhaps this last sentence should be reformulated by replacing ‘test’ by ‘standard test’, since for true CSR patterns there are always small deviations from a Poisson process behaviour which can be used, after inspection of the given pattern, in a highly specialised test for rejecting CSR.)

In the following, a series of CSR tests will be presented. Note that some of these tests use point process characteristics which have not yet been explained. These will be introduced in Chapter 4. Glance over the text below and then go to Chapter 4 to obtain the necessary information and return to this chapter.

The CSR tests are presented in order of data quality, from the coarsest to the most detailed data type:

- quadrat counts,
- distance measurement,
- measurement of directions to neighbours as angles,
- use of mapped data.

When the point pattern has been mapped exhaustively, such that measurements of the locations of all points in the pattern are available, it is worthwhile to use more sophisticated statistics and tests. These rely on computers and point process statistics software. Two approaches have been devised.

- The traditional approach uses point process statistics software, for example, for the estimation of the L - or J -function.
- The more statistical approach is based on the fact that the point distribution of a Poisson process is uniform. If the window W is a rectangle or a parallelepiped then it can be transformed to the unit square or cube and the given point pattern can be considered a sample from a binomial process or even a periodic binomial process. It is then possible to use discrepancy measures from quasi-Monte Carlo methods in numerical integration techniques (Hua and Wang, 1981; Niederreiter, 1992; Hickernell et al., 2005), which evaluate the discrepancy between the empirical point distribution and the uniform distribution; see Ho and Chiu (2007a). Alternatively, the usual summary characteristics of point processes under the periodic binomial process assumption are estimated with an adapted edge-correction; see Section 2.7.7.

A quite natural approach is the application of goodness-of-fit tests for the uniformity hypothesis. This includes classical goodness-of-fit tests, such as the χ^2 -test in Section 2.7.2 and the Kolmogorov–Smirnov test in Section 2.7.7. The authors' favourite test is the L -test based on the L -function, which requires mapped data. For the sake of the exposition, the two-dimensional case is mainly considered here. In most cases a generalisation to the three-dimensional case is straightforward.

General idea of CSR tests

Most CSR tests are constructed as follows. A summary characteristic is estimated for the data and compared with the relevant theoretical summary characteristic for a Poisson process. If there is a large difference between both characteristics, the Poisson null hypothesis is rejected. The tests may be based on either numerical summary characteristics, i.e. a single value, or functional summary characteristics, i.e. a function of distance r .

Numerical summary characteristic M : Let $\hat{M}$ be the estimator of some summary characteristic M and M_p the theoretical value for a Poisson process. Reject the CSR hypothesis if

$$|\hat{M} - M_p| > M_\alpha,$$

where M_α is some critical value. The test is more powerful if it depends on the alternatives ‘clustering’ and ‘regularity’: reject the CSR hypothesis if

$$\hat{M} > M_\alpha \quad \text{if clustering alternative}$$

and

$$\hat{M} < M_{1-\alpha} \quad \text{if regularity alternative.}$$

An example of such a test is the mean-direction test; see Section 2.7.5. Note that the terms ‘clustering’ and ‘regularity’ should not be used naively, since the behaviour of a point process can be different at different spatial scales.

Functional summary characteristic $S(r)$: Let $\hat{S}(r)$ be an estimator of some summary characteristic $S(r)$ and let $S_p(r)$ be its theoretical Poisson counterpart. Reject the CSR hypothesis if

$$\max_{r \leq s} |\hat{S}(r) - S_p(r)| > S_\alpha^m \quad (2.7.1)$$

or if

$$\int_0^s (\hat{S}(r) - S_p(r))^2 dr > S_\alpha^i. \quad (2.7.2)$$

The test statistics in (2.7.1) and (2.7.2) are called the *maximum statistic* and *integral statistic*, respectively (Thönnies and Van Lieshout, 1999). In the integral of (2.7.2), $(\cdot)^2$ can be replaced by $|\cdot|$ or $|\cdot|^\gamma$ with some $\gamma > 0$.

The choice of the maximum r -value s is crucial. Advice on the appropriate choice of s is given for the specific tests. Choosing values for s that are too large reduces the power of the tests. This is because r is usually an (inter-point) distance and, for large r , $S(r)$ cannot be estimated from a bounded window of observation or only with large variance.

Finally, the method used for the calculation of $\hat{S}(r)$ has an impact on the result. This clear fact was ignored for a long time but demonstrated by Ho and Chiu (2006) for the L -test.

In some cases, critical values for CSR tests are known. If this is not the case, simulation tests have to be used. A test of this type is discussed on p. 96 in the context of the L -test.

The alternative hypotheses are usually *clustering* or *regularity*, and it is assumed that background knowledge suggests which alternative is most suitable in the given context. Indeed, this assumption is realistic, since some anecdotal evidence or prior knowledge is typically available: for example, in forestry it is known that young trees in natural forests tend to show clustering as the seedlings sprout in proximity to the parent tree, while old trees in cultivated forests tend to be regular. In materials science applications, knowledge of the processes that lead to the structures enables a prediction as to whether the alternative hypothesis should be one of regularity or clustering. Visual inspection of the pattern is often helpful.

The alternative hypothesis has to be chosen prior to data collection. It is not a good idea to estimate a summary characteristic, to establish the alternative hypothesis based on a comparison with the corresponding Poisson process value and to then show that the observed deviation is significant.

There are, of course, other types of deviation from the CSR hypothesis: the point pattern may show clustering and regularity in combination, for example at different spatial scales or in different regions. Also, *inhomogeneity* or non-stationarity is an important case of non-CSR. Often inhomogeneity causes CSR tests to indicate (spurious) clustering.

Example 2.2. Gold particles: CSR testing

In the following, the pattern of gold particles introduced in Chapter 1 (see Figure 1.3) will serve as a test pattern and all CSR tests will be applied to it. Visual inspection encourages this choice: the pattern looks rather homogeneous, hence the assumption that it is a sample of a stationary and isotropic point process can be accepted.

The following will reveal the results based on the statistical methods for the gold pattern, eventually yielding a clear result. Prior to reading this, readers might now take a moment to write down what they think of this pattern after visual inspection, and why.

Figures 4.14, 4.18 and 4.20 may be consulted which show the empirical nearest-neighbour distance d.f., the L -function and the pair correlation function in comparison to the corresponding Poisson process characteristics. Figure 4.18 shows the estimated L -function for the gold particle pattern and the maximum and minimum envelopes from 99 simulations of binomial processes with 218 points in a 630×400 window. There are big deviations for small r (up to $r = 5.65$) because of the hard-core of the empirical pattern. Note also that $\hat{L}(r)$ is close to r at distances as close as $r = 30$, which may mean that there is some clustering, combined with the hard-core effect.

The pattern of positions of *Phlebotomus* in Figure 1.4 shows such a high degree of clustering that a CSR does not make sense; it cannot be regarded as a sample of a homogeneous Poisson process.

2.7.2 Quadrat counts

The following tests are suitable if the sampling window W has been divided into k different subregions of equal area $\nu(Q)$. In this case, the point pattern has been sampled as counts of numbers of points falling into these defined subregions. In

applications, these are often *quadrats* or squares, i.e. the data have been collected on a grid with rectangular grid cells of equal size. Note that mapped data may be transformed into quadrat counts by admitting a grid on the pattern, and the following tests are equally applicable then. However, since this results in a loss of valuable information, the use of the better adapted methods described below (e.g. Section 2.7.7) is recommended.

Of course, the index-of-dispersion test and quadrat count tests in general have the disadvantage that quite different processes may yet yield quadrat counts of a similar distribution. Furthermore, the test depends on the number k of quadrats, which can be chosen freely.

Under the hypothesis of a homogeneous Poisson process, the random number of points counted in these quadrats follows a Poisson distribution of mean $\lambda \cdot \nu(Q)$ and counts in disjoint quadrats are independent. This is convenient, as statistical tests can be based on these distributional properties and it is not necessary to apply simulation-based tests.

First the index-of-dispersion test is introduced, which is the simplest test. It is followed by its refined variant, the Greig-Smith method, which investigates the extent to which the independence properties are valid.

Index-of-dispersion test

The index of dispersion I is defined by

$$I = \frac{(k-1)s^2}{\bar{x}}, \quad (2.7.3)$$

where k is the number of quadrats, $\bar{x}$ is the mean number of points per quadrat, and s^2 is the sample variance of the number of points per quadrat.

This test statistic may seem to have been inspired by the fact that the mean and variance of a Poisson distribution are equal. But this is not the case: it is exactly the same test statistic as that of a χ^2 goodness-of-fit test of the hypothesis that the n points are independently and uniformly distributed in W . Consequently, the index I follows approximately a χ^2 -distribution with $k-1$ degrees of freedom provided that $k > 6$ and $\lambda \cdot \nu(Q) > 1$ (Diggle, 2003).

Hence, if I exceeds $\chi_{k-1;\alpha}^2$ or I is smaller than $\chi_{k-1;1-\alpha}^2$ the test rejects the CSR hypothesis at a significance level of $100\alpha\%$.

In the first case, the alternative hypothesis is that the variability in the process is stronger than for the Poisson process, i.e. that there is aggregation. Analogously, in the second case the alternative hypothesis is that there is some regularity in the point pattern as the variability is smaller.

Table 2.5 Results of the index-of-dispersion test for the gold particles.

k	$\chi^2_{0.95}$	I	$\chi^2_{0.05}$
4	0.352	0.46	7.82
16	7.26	13.05	25.0
64	46.6	70.88	82.5
256	218.7	319.83	293.0

Table 2.6 Quadrat counts for the gold particles and $k = 64$.

3	4	6	2	8	4	0	1
5	1	2	4	2	2	5	2
6	1	3	6	2	7	4	2
3	2	2	7	5	4	3	6
4	4	1	3	4	3	0	6
2	6	2	3	3	8	3	1
2	1	2	2	3	2	6	4
4	3	6	7	2	3	2	2

Example 2.3. Gold particles: index-of-dispersion test

Now the index-of-dispersion test is applied to the gold particle data. In addition to illustrating its use, the test's heavy dependence on the choice of grid can be illustrated well with this example.

Table 2.5 shows the results of the index-of-dispersion test for $k = 4, 16, 64$ and 256. For the smaller k , i.e. the larger quadrats, the CSR hypothesis is not rejected, which might show that on a larger scale the pattern behaves like a Poisson process. However, for $k = 256$ the Poisson hypothesis is clearly rejected – at small distances there are deviations from Poisson behaviour, i.e. clustering.

The quadrat counts for the case of $k = 64$ are shown in Table 2.6. Here the 'quadrats' are rectangles of side lengths $630/8$ and $400/8$.

Greig-Smith method

This method, introduced in Greig-Smith (1964), is an improvement on the simple quadrat count as it does not consider a single quadrat size but also uses counts grouped by neighbouring quadrats. In this way, clustering at different scales may be detected. The example below illustrates this.

It is assumed that the number of quadrats is $k = 2^q$ for some integer q . The quantities $s_1, s_2, \dots$ are calculated by

$$s_1^2 = \sum_{(j)} \left(\frac{\text{number in the } j\text{th quadrat}}{\text{pair of quadrats}} \right)^2 - \frac{1}{2} \sum_{(k)} \left(\frac{\text{number in the } k\text{th}}{\text{pair of quadrats}} \right)^2,$$

$$s_2^2 = \sum_{(j)} \left(\frac{\text{number in the } j\text{th}}{\text{pair of quadrats}} \right)^2 - \frac{1}{2} \sum_{(k)} \left(\frac{\text{number in the } k\text{th}}{\text{quartet of quadrats}} \right)^2,$$

and so on.

For $q = 6$, the quadrats can be numbered in chessboard notation, since $2^q = 64$. The pairs of quadrats are then the pairs $(a1, a2), (a3, a4), \dots, (b1, b2), (b3, b4), \dots$. The quartets of quadrats are then $(a1, a2, b1, b2), (a3, a4, b3, b4), \dots$.

The quantities $s_j^2/2^{q-j}$ are unbiased estimators of the variance of the random number of points in a small quadrat. Thus, the index-of-dispersion formula may now be generalised, the indices

$$I_j = s_j^2/\bar{x}$$

may be calculated and the χ^2 -test applied to each of the indices.

Here $\bar{x}$ is the same as in the index-of-dispersion test. The degrees of freedom for the χ^2 -statistics $I_1, I_2, \dots$ are $2^{q-1}, 2^{q-2}, \dots$.

The results of the tests should be interpreted as follows. The hypothesis of a homogeneous Poisson process can be accepted if all the indices I_j lie between two-sided critical values of the appropriate χ^2 -statistics. Otherwise, the values of the indices indicate the nature of the deviation from the hypothesis. Thus if, for example, $I_3/2^{q-3}$ is significantly greater than one, there is evidence of clustering on the scale of groups of four quadrats, etc.

2.7.3 Distance methods

In some circumstances, the measurements most readily available are distances between pairs of points of the pattern, and between points of the pattern and chosen test locations. For example, this is often the case when point patterns of forest tree locations are being investigated. Distance methods take these limitations into account and work with these measurements alone. However, when the point pattern has been mapped, better tests are available, i.e. those using L - and J -functions.

Equations (2.3.6) and (2.5.8) for the distance d.f.s of the Poisson process form the basis of distance methods as applied to testing the CSR hypothesis. Deviations of the estimators $\hat{D}(r)$ and $\hat{H}_s(r)$ from $1 - \exp(-\pi\lambda r^2)$ indicate deviations from CSR. A further starting point is equation (2.5.9)

$$D(r) = H_s(r), \quad \text{for } r \geq 0.$$

This equation asserts that the nearest-neighbour distance distribution function $D(r)$ and the spherical contact distribution function $H_s(r)$ coincide for a homogeneous Poisson process. One might hope to obtain empirical distribution functions corresponding to D and H_s based on distance measurements and then test for their equality. For example, Diggle (2003) recommends the test statistic

$$\int_0^t \left(\hat{H}_s(r) - \hat{D}(r) \right)^2 dr$$

for some t , which may play the role of M on p. 85. The critical values M_α can be determined by simulation; only large deviations are relevant.

A particular difficulty arises in the distance approach: in order to obtain nearest-neighbour distances one has to sample the points of the pattern at random. However, this should be done without enumerating all points of the pattern as otherwise distance methods lose their practical advantage. How can the points of the pattern be sampled at random without a bias towards points occurring in particular locations?

Byth and Ripley (1980) give an example of a method that overcomes this problem. A number $2m$ of locations are chosen in the window W by semi-systematic sampling. Half of these are used as test points from which to measure distances to nearest points in the pattern. These distances are denoted by $u_1, \dots, u_m$. The other half of the locations are used to define subregions within which the point pattern is surveyed exhaustively. A common size for all subregions is chosen, with an expected number of about five points of the pattern falling into each subregion. A point in the pattern is selected at random from each subregion and the nearest-neighbour distance is measured for each of these points. This gives rise to m further distances $v_1, \dots, v_m$. Under the CSR hypothesis (since then $D(r) = H_s(r)$) these two samples of m numbers each are drawn from the probability distribution given by

$$F(r) = 1 - \exp(-\lambda \pi r^2) \quad \text{for } r \geq 0,$$

(ignoring edge effects). The samples are approximately independent and independent of each other, if m is not too large. Byth and Ripley (1980) suggest that m should be around 5 % of the total number of points in the window W . As a consequence of the CSR hypothesis the quantities $u_1^2, \dots, u_m^2, v_1^2, \dots, v_m^2$ are approximately independent and exponentially distributed with parameter $\lambda \pi$. Thus the statistics

$$h_F = \sum_{i=1}^m u_i^2 \bigg/ \sum_{i=1}^m v_i^2 \tag{2.7.4}$$

and

$$h_N = \frac{1}{m} \sum_{i=1}^m u_i^2 / (u_i^2 + v_i^2) \quad (2.7.5)$$

are approximately F -distributed with $(2m, 2m)$ degrees of freedom (h_F) and normally distributed with mean $1/2$ and variance $1/(12m)$ (h_N), respectively. The measurements u_i, v_i are paired for the formula involving h_N ; this pairing may be made at random.

These approximate distributional results allow the construction of statistical tests. Excessively large or small values for either h_F or h_N indicate departures from the CSR hypothesis. Which statistic is appropriate depends on the alternative hypothesis one has in mind. If the alternative involves clustering, h_F is more suitable, whereas h_N is preferred if the alternative assumes regularity in the pattern.

Details of these two tests and comparisons with other tests can be found in Byth and Ripley (1980) and Holgate (1965). For other references to distance methods, see Diggle (2003).

2.7.4 The J -test

The J -test uses distances but requires mapped data. It is based on the J -function, which is given by

$$J(r) = \frac{1 - D(r)}{1 - H_s(r)}. \quad (2.7.6)$$

For a Poisson process $J(r) \equiv 1$.

From (2.7.1) and (2.7.2) above, the test statistics are

$$\tau_{\max} = \max_{r \leq s} |\hat{J}(r) - 1|$$

and

$$\tau_{\text{int}} = \int_0^s (\hat{J}(r) - 1)^2 dr.$$

The J -function has the advantage of measuring both the strength and range of interaction and of allowing a simple interpretation; see Section 4.2.7.

This test has been thoroughly investigated by Thönnies and Van Lieshout (1999). They show that its power is strongly influenced by the choice of maximum distance s : for large values of r , $1 - H_s(r)$ is close to zero and thus $J(r)$ can have large fluctuations (which makes the J -test slightly difficult). Therefore, they recommend values of s close to the interaction radius of the point process or, more simply, for

which $H_s(r)$ is ‘sufficiently below 1’. The authors of the present book recommend applying several values of s and comparing the results.

Critical values for this test are obtained by simulation.

2.7.5 Two index-based tests

The *Clark–Evans test* is based on a numerical summary characteristic, the Clark–Evans index CE ; see Section 4.2.4. It compares the mean nearest-neighbour distance of a given pattern with the mean nearest-neighbour distance of a Poisson process of the same intensity as a given process. The estimation of CE is described on p. 198. Mapped data are not required; it suffices to measure nearest-neighbour distances.

For a Poisson process the mean value of CE is 1. Reject the CSR hypothesis if

$$CE > CE_\alpha \qquad \text{for clustering alternative}$$

or

$$CE < CE_{1-\alpha} \qquad \text{for regularity alternative.}$$

Table 2.7 gives some critical values CE_β , for $\beta = \alpha$ or $\beta = 1 - \alpha$, for a square window. These only depend on the number of points n . The values for rectangular windows are slightly smaller for α (larger for $1 - \alpha$) since edge effects are stronger and cause more variability.

Statistical experience shows that this test is not very powerful for regular patterns.

The *mean-direction test* is based on the numerical summary characteristic mean-direction index $\bar{R}_4$; see p. 197. It uses the length of the sum of the unit vectors pointing from the typical point to its four closest neighbours. The estimation of $\bar{R}_k$ is described on p. 199. Mapped data are not required; measurement of the angles for the directions from the points to their neighbours is sufficient.

Table 2.7 Critical values CE_β by number of points n for a square window.

β	n					
	25	50	100	200	400	600
0.99	0.58	0.72	0.80	0.86	0.90	0.92
0.95	0.68	0.77	0.84	0.89	0.93	0.94
0.05	1.12	1.06	1.04	1.03	1.02	1.01
0.01	1.22	1.11	1.08	1.05	1.03	1.03

Table 2.8 Critical values $\overline{R}_{4,\beta}$ as a function of number of points n for a square window.

β	n					
	25	50	100	200	400	600
0.99	0.90	1.31	1.50	1.60	1.65	1.68
0.95	1.14	1.43	1.57	1.66	1.69	1.72
0.05	2.40	2.16	2.04	1.94	1.90	1.87
0.01	2.73	2.33	2.11	2.00	1.95	1.91

For a Poisson process the theoretical value of $\overline{R}_4$ is 1.799. This was obtained by simulation from 10^6 quadruples of unit vectors.

The CSR hypothesis is rejected if

$$\overline{R}_4 > \overline{R}_{4,\alpha} \quad \text{for clustering alternative}$$

or

$$\overline{R}_4 < \overline{R}_{4,1-\alpha} \quad \text{for regularity alternative.}$$

Again, this test shows only weak reactions against hard-core distances in the pattern.

Table 2.8 gives some critical values $\overline{R}_{4,\beta}$, $\beta = \alpha$ and $\beta = 1 - \alpha$, for a square window, which only depend on the number n of points (Corral-Rivas, 2006). The influence of the shape of the window is small.

2.7.6 Discrepancy tests

If the pattern has been sampled in a rectangular window, it makes sense to test the CSR hypothesis by means of classical methods of statistics, by comparison of the two-dimensional empirical distribution function $U_n(x)$ of the uniform distribution on $\mathbb{I}$ with its theoretical counterpart $U(x)$. For this purpose the original points $x_i = (\xi_{i,1}, \xi_{i,2})$ ($0 \leq \xi_{i,1} \leq a$, $0 \leq \xi_{i,2} \leq b$) are transformed to the unit square $\mathbb{I}$ by $\eta_{i,1} = \xi_{i,1}/a$ and $\eta_{i,2} = \xi_{i,2}/b$. Then

$$U(\eta) = \eta_1 \eta_2$$

and

$$U_n(\eta) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(\eta_i < \eta),$$

for $\eta = (\eta_1, \eta_2)$ and $\eta_i = (\eta_{i,1}, \eta_{i,2})$. A suitable test statistic is the L^2 -star discrepancy

$$\omega^2 = \int_0^1 \int_0^1 (U(\eta_1, \eta_2) - U_n(\eta_1, \eta_2))^2 d\eta_1 d\eta_2.$$

This test characteristic ω^2 is ‘flawed because its value depends on which corner of W one chooses as the origin of the co-ordinate system’ (Zimmerman, 1993). Therefore, Zimmerman decided to choose the average of analogous characteristics for all four corners, which leads to the test characteristic

$$\begin{aligned} \bar{\omega}^2 &= \frac{1}{4n} \sum_{i=1}^n \sum_{j=1}^n (1 - |\eta_{1,i} - \eta_{1,j}|)(1 - |\eta_{2,i} - \eta_{2,j}|) \\ &\quad - \frac{1}{2} \sum_{i=1}^n \left(\eta_{1,i}^2 - \eta_{1,i} - \frac{1}{2} \right) \left(\eta_{2,i}^2 - \eta_{2,i} - \frac{1}{2} \right) + \frac{n}{9}. \end{aligned}$$

A large value of $\bar{\omega}^2$ indicates deviations from uniformity such as clustering or heterogeneity, while small values indicate that the pattern is more regular. Zimmerman (1993) gives the critical values for $n > 20$ shown in Table 2.9.

If $\bar{\omega}^2$ exceeds ω_α (if the alternative is clustering or inhomogeneity) or is smaller than $\omega_{1-\alpha}$ (if the alternative is regularity) the test rejects the CSR hypothesis at significance level $100\alpha\%$. According to Zimmerman (1993), this test is particularly interesting for patterns which might be inhomogeneous. Ho and Chiu (2007a) consider other discrepancies, or measures of uniformity, that are also independent of the choice of origin, and show that their discrepancies are powerful statistics against not only inhomogeneous but also clustered alternatives. However, neither Zimmerman’s nor Ho and Chiu’s discrepancies are very successful at detecting regularity.

The ideas can be easily extended to the general d -dimensional case.

Table 2.9 Critical values ω_β of the discrepancy test.

β	ω_β
0.99	0.043
0.98	0.049
0.95	0.057
0.05	0.281
0.02	0.342
0.01	0.389

2.7.7 The L -test

The L -test is based on the fact that the L -function of a Poisson process has the simple linear form

$$L(r) = r \quad \text{for } r \geq 0.$$

Deviations of empirical L -functions from $L(r) = r$ can be used to test the CSR hypothesis. It is indeed better to use the L -function and not the K -function, which also has a simple form,

$$K(r) = \pi r^2.$$

The reason is that the square root transformation

$$L(r) = \sqrt{K(r)/\pi} \quad \text{for } r \geq 0$$

stabilises variance as discussed in Section 4.3.1.

It is then natural to use the test statistics

$$\tau = \max_{r \leq s} |\hat{L}(r) - r| \quad (2.7.7)$$

with

$$\hat{L}(r) = \sqrt{\hat{K}(r)/\pi},$$

where $\hat{K}(r)$ is some estimator of the K -function and s is a suitable maximum distance.

It is advisable to run the L -test based on standard point process software, such as the `spatstat` library in R; see Baddeley and Turner (2005, 2006). Good choices for $\hat{K}(r)$ are Ripley's isotropic estimator of the K -function (see Section 4.3),

$$\hat{K}(r) = \hat{\kappa}_R(r)/\hat{\lambda}^2,$$

or the stationary estimator

$$\hat{K}(r) = \hat{\kappa}_{st}(r)/\hat{\lambda}_v(r)^2.$$

If τ is large, then the CSR hypothesis is rejected. (The alternative hypothesis is 'no Poisson process' without further specification.) The critical value of τ for the significance level $\alpha = 0.05$ is approximately

$$\tau_{0.05} = 1.45\sqrt{a/n}, \quad (2.7.8)$$

where $a = \nu(W)$ is the window area and n is the number of observed points (Ripley, 1988, p. 46). This value was obtained by simulations and can be used if nus^3/a^2 is small for a ‘wide range of’ maximum distances s . Here, $u = U(W)$ is the boundary length of W . For $\alpha = 0.01$ the analogous value is

$$\tau_{0.01} = 1.75\sqrt{a}/n; \quad (2.7.9)$$

see Chiu (2007). So far statisticians have been unable to determine the exact critical values analytically.

If Ripley’s approximation appears to be inappropriate, simulations may be used to provide a *Monte Carlo test*. Here k (e.g. $k = 999$) independent samples from a binomial process with n points are simulated in the window W . For each of the samples the L -function is estimated and the value

$$\tau_i = \max_{r \leq s} |\hat{L}_i(r) - r| \quad \text{for } i = 1, 2, \dots, k,$$

is determined. These values and the corresponding value τ for the empirical data are ranked in ascending order. If the rank of τ exceeds $(1 - \alpha)(k + 1)$, the CSR hypothesis is rejected for the error probability α . The position $n(\tau)$ of τ gives an approximation of the p -value; it is around $1 - n(\tau)/(k + 1)$.

A simplified test uses the minimum inter-point distance $\hat{r}_0$ in the pattern (Ripley and Silverman, 1978). If this is too large, the CSR hypothesis may be rejected and some regularity alternative accepted. The rule is: reject CSR if

$$\hat{r}_0 > c\sqrt{a}/n,$$

with $c = 1.38$ for $\alpha = 0.05$ and $c = 1.71$ for $\alpha = 0.01$ for the planar case and $c = 1.20$ for $\alpha = 0.05$ and $c = 1.48$ for $\alpha = 0.01$ for the spatial case.

In the literature, different recommendations have been given for the choice of the maximum distance s of r . Ripley (1979) suggests that s should be $0.25l$ for $n = 25$ and $0.125l$ for $n = 100$ if W is a square of side length l , while Diggle (2003) suggests that s should not be bigger than $0.25l$. Ho and Chiu (2006) demonstrate that values of s in the order of $\hat{r}_0$ or of the mean cluster diameter also perform well. However, the same authors show that if the L -function is estimated by $\hat{\kappa}_{st}(r)$ and $\hat{\lambda}_v^2(r)$ (see Section 4.2.3) the variance-stabilising effect of the use of the square-root transformation $\hat{K} \rightarrow \hat{L}$ in combination with the adapted intensity estimator is so strong that the power of the L -test is nearly independent of s , and so half the diagonal length can be safely recommended.

According to Ho and Chiu (2007b) the L -test can be improved by introducing weight functions $w(r)$, which reduce the importance of very small or large r ,

$$\tau = \max_{r \leq s} \{|\hat{L}(r) - r| \cdot w(r)\}.$$

They show that decreasing functions such as $w(r) = 1 - \frac{2r}{l}$ or $w(r) = 10^{-r/l}$ are helpful in the case of processes with short-range correlation, while weight functions such as $w(r) = \frac{r}{l} e^{-r/l}$ seem to be suitable for processes with long-range correlation; l is the side length of the square window as above. They suggest using the empirical pair correlation function $\hat{g}(r)$ to determine whether the range of correlation may be long or short before applying their approach.

2.7.8 Other tests and recommendations

Many other CSR tests have been developed and used; some of them are mentioned here for the reader's convenience.

Instead of the L - or the J -function, $H_s(r)$ and $D(r)$ were used in (2.7.2) as functional summary characteristics; see Diggle (1979), Thönnies and Van Lieshout (1999) and Myles et al. (1995). The latter paper uses the d.f. of the cell area for the Dirichlet tessellation with respect to the points in the sample. Since this function is not known analytically, it was determined by simulation; see also Chiu (2003). Mecke and Stoyan (2005) use the morphological function $n(r)$ (see p. 202) for a CSR test, which seems to have power comparable to that of the L -test. A more sophisticated test is the Q^2 -test of Grabarnik and Chiu (2002), which uses the numbers of points with k neighbours within a given distance r . The length distribution of the minimal spanning tree is used in Hoffman and Jain (1983). The basic idea is that the minimal spanning tree tends to be shorter for a cluster process and longer for a regular process, in comparison to the CSR case. Spectral tests are discussed in Muggleston and Renshaw (2001).

So far, no one has systematically compared all CSR tests. Myles et al. (1995) compare the K - and L -test with the index-of-dispersion test and the tests based on $D(r)$ and Dirichlet cell area distribution and favour the L -test. They use the Thomas cluster process and the RSA process as alternative models; see Section 6.3.2 and 6.4.4. Grabarnik and Chiu (2002) compare tests based on their summary characteristic Q^2 , on $D(r)$, $K(r)$, the reduced third moment function and on the variance–area curve. Their results are rather complex and show that there is no uniformly best test.

Recommendations

- If you have mapped data and do not want to apply a simulation test, then use the L -test with (2.7.8) and (2.7.9).
- If you can carry out a simulation test then use the L -test based on the periodic binomial process. Compare the results with that of a classical L -test and a J -test.
- If you have only quadrat counts as data or if you want to work without a computer, for instance in the field, use the index-of-dispersion test.
- If you do not have a map of the points but you can visit each point, then use the CE -test and the mean-direction test.

Example 2.4. Gold particles: advanced CSR testing

The more advanced tests described in Sections 2.7.4–2.7.7 yield the following results.

J-test. The test was carried out with the maximum statistic and estimates of the J -function constructed by means of edge-corrected estimates of $D(r)$ and $H_s(r)$. For the distance s the values $s = 8, 10$ and 20 were used in parallel. For $s = 8$ a P -value of 0.058 was obtained by simulation, but for $s = 10$ and 20 the P -values were very small (less than 0.0001) and 0.009 , respectively. Thus, the CSR hypothesis is not rejected for $s = 8$ but is rejected for the larger values of s . The reader should consider Figures 4.17 and 4.20 in Chapter 4 in order to understand the choice of the values of s .

Mean-direction test. The empirical mean-direction index $\bar{R}_4$ is 1.9095 , i.e. between the critical values for $n = 200$ and $\alpha = 0.05$; the CSR hypothesis is not rejected.

Discrepancy test. The test statistic $\bar{\omega}^2$ results in a value of 0.0766 and lies between the limits 0.057 and 0.281 . This test does not reject the CSR hypothesis either.

L-test. The test statistic is $\tau = 5.65$. This value results from the minimum inter-point distance of 5.65 in the pattern, which implies $\hat{L}(5.65) = 0$. The critical value $\tau_{0.05}$ is

$$\tau_{0.05} = \frac{\sqrt{650 \cdot 400}}{218} = 3.39,$$

and so the Poisson hypothesis is rejected. The choice of s does not make any difference to the test results for the gold particle pattern. The test based on $\hat{r}_0$ already rejects the CSR hypothesis.

In their original analysis of the gold particle pattern, Glasbey and Roberts (1997) also reject the CSR hypothesis, based on a test using

$$\int_0^s \left(\hat{D}(r) - (1 - \exp(-\lambda \pi r^2)) \right)^2 dr.$$

In summary, the gold particle pattern should not be regarded as a sample from a homogeneous Poisson process. It can perhaps be considered as a point pattern which is globally similar to a Poisson process, but locally there are heavy deviations because of the hard-core distance of $\hat{r}_0 = 5.65$ and because of small-scale clustering.

The behaviour of the tests when applied to the data set in Example 2.4 may support the prejudgemental idea that the more complicated tests are also more powerful than the simple discrepancy and index-based tests. But the reader should note that the authors have experienced cases where the L -test confirms CSR and the mean-direction test rejects it. It is always useful to study the nature of a given pattern and to identify a summary characteristic that indicates well any deviations from CSR and then to use a test based on exactly that summary characteristic.